CHAPTER 2 · CONCEPT 4 · WORKSHEET

Standard Form — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ **[1]**

The standard form of 59000 is:

(a) 5.9×10^3 (b) 5.9×10^4 (c) 59×10^3 (d) 0.59×10^5

Q2 · MCQ **[1]**

0.00037 in standard form is:

(a) 3.7×10^{-4} (b) 3.7×10^4 (c) 37×10^{-5} (d) 3.7×10^{-3}

Q3 · FILL IN THE BLANK **[1]**

In standard form $x \times 10^y$, the value of x must be at least 1 and less than:

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Q4 · MCQ **[1]**

3.081×10^8 written as an ordinary number is:

(a) 30810000 (b) 308100000 (c) 3081000000 (d) 30810000000

Q5 · TRUE / FALSE **[1]**

A very small number such as 0.0004 has a negative power of 10 in standard form.

True False

Q6 · ASSERTION–REASON [1]

Assertion (A): $5900 = 5.9 \times 10^3$.
Reason (R): The standard form of a number is $x \times 10^y$, where $1 \le x < 10$ and y is an integer.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7 [2]

Write in standard form: (i) 39,00,00,000 (ii) 0.000005678.

Q8 [2]

Express the product $3.2 \times 10^6 \times 4.1 \times 10^{-1}$ in standard form.

Q9 [2]

Write 47561 in expanded form using powers of 10.

Long answer (3 marks each)

Q10 [3]

Write in standard form: (i) 1,49,60,00,00,000 (ii) 0.000000123.

Q11 **[3]**

Planet A is at a distance of 9.35×10^6 km from Earth and planet B is 6.27×10^7 km from Earth. Which planet is nearer to Earth? Explain.

Q12 · CASE STUDY **[3]**

A bacteria culture starts with 2^3 cells and doubles every hour.
(a) How many cells are there after 2 hours?
(b) Express the number after 5 hours in exponential (power of 2) form.
(c) Write that answer as an ordinary number.

Total: 21 marks · Check your work against the answer key.